

Kathirvel Rajmohan

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Professional Summary

Data Engineer with around 2 years of experience writing complex SQL and building PySpark data pipelines in Databricks — including cluster configuration, job orchestration, and AWS S3 integration — alongside enterprise application development. Databricks Certified Data Engineer Associate, with additional hands-on experience in Delta Lake, data quality frameworks, and Medallion architecture through an independent end-to-end ETL project.

Skills

Languages: Python, SQL, Java, JavaScript, C/C++
Data Engineering: Apache Spark (PySpark), Databricks (Jobs, Workflows), Delta Lake, ETL/ELT Pipeline Design, Data Quality & Validation, Star Schema / Dimensional Modeling, Medallion Architecture (Bronze/Silver/Gold)
Databases: PostgreSQL, MySQL, MongoDB, Spark SQL
Tools & Other: Git, REST APIs, Data Structures & Algorithms, Agile/Scrum

Certifications

Databricks Certified Data Engineer Associate 2026
Validates expertise in lakehouse architecture, Delta Lake, ETL pipelines, Apache Spark, and data engineering best practices.

Work Experience

Optimum Solutions, Chennai Oct 2024 – Present
Associate Consultant

- Wrote and optimized **complex SQL queries** involving multi-table joins, CTEs, aggregations, and conditional logic to support application data operations, validation, and reporting.
- Built and tested **Python/PySpark data pipeline notebooks in Databricks**, configuring compute **clusters** and scheduling jobs via **Databricks Workflows**, and reading/writing data to/from **AWS S3** for downstream reporting and analytics.
- Identified a performance bottleneck in a SQL-based transformation processing 1M+ rows (50-minute runtime); redesigned it in PySpark, cutting runtime to 13 minutes (**~74% faster**) and improving pipeline turnaround.
- Developed and delivered new features, batch processes, and bug fixes across multiple project repositories using **Git** for version control and branch-based development workflows.
- Performed **unit testing (UT)** and **functional testing (FT)** to validate feature correctness and ensure quality before production deployment.
- Collaborated with cross-functional teammates and stakeholders in an **Agile/Scrum** environment, providing daily stand-up updates and aligning on sprint goals.

Education

IFET College of Engineering, Villupuram, Tamilnadu Sep 2021 – Apr 2024
B.E. in Computer Science and Engineering **CGPA: 8.4/10**
Relevant Coursework: Data Structures and Algorithms, Database Management Systems, Data Warehousing and Data Mining, Machine Learning, Object Oriented Programming, Operating Systems, Computer Networks

Seshasayee Institute of Technology July 2018 – Jun 2021
Diploma in Electrical and Electronics Engineering **CGPA: 8.2/10**
Relevant Coursework: C++ Programming, Programmable Logic Controller, Logic Gates

Project Work

- Sales ETL Pipeline — Databricks, PySpark, Delta Lake (Ongoing):** Building an end-to-end data engineering pipeline on Databricks using Medallion Architecture (Bronze/Silver/Gold) on a retail sales dataset. Implemented Delta Lake tables with ACID transactions and schema evolution, PySpark DataFrame transformations and window functions, and a 53-check automated data quality validation framework with severity-based pipeline gating. Modeled a star schema (fact and dimension tables) and scheduled the pipeline via Databricks Jobs with failure alerting. Currently extending with AWS S3 integration, Docker, CI/CD, and Airflow orchestration.
Tools and Technology Used: Python, PySpark, Databricks, Delta Lake, Spark SQL, Star Schema Modeling, Databricks Jobs.